MILITARY SPECIFICATION

EXCAVATOR, MULTIPURPOSE, WHEEL-UNDERCARRIAGE, DIESEL-ENGINE-DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an undercarriage, revolving upperstructure, one diesel engine, two-piece boom type multipurpose excavator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting.

MILITARY

MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.
MIL-C-3580 - Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and Their Attachments, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3805

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STANDARD

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

2.1.2 Other Government documents. The following Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

POWER CRANE AND SHOVEL ASSOCIATION (PCSA)

PCSA Standard No. 5 - Mobile Hydraulic Excavator Standards.

(Application for copies should be addressed to the Power Crane and Shovel Association, Marina Plaza, Suite 1700, 111 E. Wisconsin Avenue, Milwaukee, WI 53202-4879.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The excavator shall consist essentially of a revolving upperstructure, operating a bucket attached on the end of a boom assembly, mounted on a pneumatic-tired carrier. The revolving upperstructure shall consist of an operator's cab, all operational controls, hydraulic system, and the boom assembly. The diesel engine shall power the hydraulic system, provide motivation for all functions including travel function.

3.2 First article. When specified (see 6.2), the contractor shall furnish an excavator for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The excavator shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the excavator being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model. As a minimum the excavator shall meet the requirements of PCSA Standard No. 5 for Mobile Hydraulic Excavators.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Maintainability. The excavator shall be designed so that the components are readily accessible for repair or replacement with minimum removal or disturbance of adjacent parts or components, using general-purpose tools. Operating parts, accessories, and drain outlets shall be readily accessible for regular maintenance service.

3.7 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperatures, that are of such nature or so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded or insulated to the extent necessary to eliminate the hazard. Walking or working surfaces and platforms shall be of an anti-skid type. Hand holds and/or steps shall be provided to allow safe entrance and exit from the operator's cab.

3.7.1 Noise. If the noise level in the area occupied by the operator exceeds 85 decibels (dB), when tested in accordance with SAE J919, a caution plate shall be permanently posted on the excavator in a conspicuous protected location and shall be clearly visible and legible to all personnel exposed to excessive noise levels. The caution plate shall read: CAUTION - HEARING PROTECTION REQUIRED. The plate shall have a yellow background with black lettering and plate shall be made of corrosion-resistant material. Exterior noise shall not exceed 88 dB(A) and be verified in accordance with SAE J88.

3.7.2 Electrical warning decal. An electrical warning decal, in accordance with OSHA requirements, cautioning the operator about operating the excavator near electric power lines, shall be provided.

3.7.3 Strike hazard. When specified (see 6.2), the rear surfaces of the excavator shall be alternating black and yellow stripes to denote a hazard to personnel on the ground.

3.8 Environmental conditions. The excavator shall start and maintain constant performance characteristics as specified herein under any of the following conditions or combination of conditions:

a. Temperature conditions. In any ambient temperature from 110 degrees Fahrenheit (°F) to -20°F, except as specified in 3.8b below.
b. Elevation conditions. At any elevation from sea level (barometric pressure 29.92 inches mercury) and a maximum ambient temperature of 110°F to 5,000 feet (ft) (barometric pressure 24.89 inches mercury) and a temperature of 100°F.

3.9 Revolving upperstructure. The upperstructure shall consist essentially of a platform complete with operator's cab, boom assembly, hydraulic asssembly, turntable, and controls mounted on a revolving platform.

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3.9.1 Cab. The cab shall be fabricated of sheet metal or sheet fiber glass and shall have sufficient windows and glazed doors to permit 270° vision for the operator. All glass shall be of the safety type, and shall be held in place by channels of rubber composition or other suitable weatherproof materials for easy replacement. Cab door(s), whether of the sliding or swinging type, shall be adequately restrained from accidentally opening or closing while the excavator is traveling or operating. A comfortable seat for the operator shall be provided. When specified (see 6.2), the cab shall be provided with a diesel-fired heater. The heater shall direct a steady flow of heated air to the operator's feet, and the defroster shall be capable of defrosting at least 75 percent of the window area. A high capacity heater used for dual purpose as a heater and a defroster combination is permissible. The defroster and heater shall be mounted and positioned to permit routine maintenance and quick removal.

3.9.2 Swing lock mechanism. A means capable of restraining the upperstructure in any desired position, with respect to the undercarriage, shall be provided. The swing lock or brake mechanism shall be in accordance with PCSA Standard No. 5 and shall be located within easy reach of the operator.

3.10 Operating controls. All functions shall be in accordance with SAE J1177 and to be controlled by the least possible quantity of hand levers and foot pedals necessary for efficient operation. The levers and pedals shall be conveniently located and arranged for easy access and operation by the operator while in a seated position and in accordance with SAE J1177. All brakes shall be capable of being easily locked or unlocked. All operating controls shall be identified in accordance with SAE J298 recommended practice.

3.11 Hydraulic system and operation. The hydraulic system shall consist of oil reservoir, pumps, full flow filtering system, manifold, hydraulic cylinders, oil temperature gauge(s), overloading and pressure relief valves, piping, tubing, high pressure hoses and connections required to properly operate the excavator. High pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working presssure. High pressure hydraulic hoses shall have the physical qualities equal to or conforming to SAE J517. Hydraulic oil shall be of the noncorrosive type with the characteristics to withstand a temperature range from -10°F to 220°F. All functions shall be controlled from the operator's position. The following minimum functions shall be hydraulically operated and controlled:

a. Boom and bucket operation.
b. Revolving of upperstructure.
c. Operation of optional attachments and accessories (see 3.12).
d. Hydrostatic drive function (see 3.14.1).

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3.12 Boom and attachments. The excavator shall be supplied with a two-piece boom with the following characteristics and capabilities when measured in accordance with SAE J1193 and SAE J1179 recommended practice.

a. Maximum radius at ground line (AA) not less than 30 ft.
b. Maximum digging depth (AB) not less than 18 ft.
c. Dumping height (BD) not less than 15 ft.
d. Swing shall be continuous.
e. Digging through a continuous range around the excavator not less than 360°.
f. Bucket digging force not less than 2200 pounds (lb).
g. Rated arm force not less than 15,500 lbs.

The following excavating attachments and accessories shall be furnished when specified (see 6.2).

		Minimum Capacity (cubic yard)
h.	Bucket, excavating, 24-inch cutting width	0.56 yd^3
i.	Bucket, excavating, 36-inch cutting width	0.88 yd^3
j.	Bucket, excavating, 48-inch cutting width	1.00 yd^3

k. Hydraulic impact hammer. The excavator shall be equipped with a hydraulic impact hammer which shall be capable of operating from the excavator's hydraulic system. The hammer and excavator shall be equipped with a quick disconnect systems to allow rapid hookup or removal of the hammer. The necessary pins, bolts, etc., shall be furnished by the manufacturer to enable hookup of the impact hammer to the excavator arm. The hammer, as a minimum, shall meet the following requirements:

(1) Energy output per minute . . . . 625,000 ft lbs minimum.
(2) Impact blow . . . . . 1,000 lbs minimum.
(3) Blows per minute . . . . . 500 minimum.

As a minimum the hydraulic impact hammer shall be furnished with the following tools:

(1) Moil point . . . . . 3 each.
(2) Chisel . . . . . 1 each.
(3) Line cutter . . . . . 1 each.
(4) Spade . . . . . 1 each.
(5) Tamper plate assembly . . . . 1 each (144 ^{2}in minimum face area).

l. Hydraulic vibratory compactor. The excavator shall be equipped with a hydraulic vibratory compactor which shall be capable of operating from the excavator's hydraulic system. The hydraulic vibratory compactor and excavator shall be equipped with quick disconnect systems to allow rapid hookup or removal. The necessary

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pins, adapters, etc., shall be furnished by the manufacturer to enable hookup of the hydraulic vibratory compactor to the excavator's arm.

The hydraulic vibratory compactor as a minimum shall have the following operating characteristics:

(1) Impulse force . . . . 20,000 lbs minimum.
(2) Cycles per minute . . 2,000.
(3) Hydraulic flow . . . . . 40 gallons per minute.
(4) Compaction plate area . . . . . 9 square ft.

The excavator shall be capable of generating the compactor plate rated impulse force of 20,000 lbs at 2,000 cycles per minute without loss of stick/boom function while operating the compactor plate.

3.13 Lifting capacity and tipping load. Lift capacities shall be calculated and in compliance with SAE J1097. Loads shall not exceed 87 percent of hydraulic lifting capacity or 75 percent of tipping capacity. A load chart for the entire digging range must be shown. With outriggers fully extended and firmly touching the contact surface, the excavator shall lift and be capable of continuous swing (360°) a load not less than 9300 lbs at 15 ft radius.

3.14 Undercarriage. The undercarriage for the excavator shall be a heavy duty frame, with not less than two axles or drive units, supported by pneumatic rubber tires, which a revolving upperstructure is mounted on. The undercarriage shall have not less than one operator's entrance step(s) to assist the operator to gain entrance to the upperstructure cab (see 3.9.1).

3.14.1 Transmission. Engine power shall be transmitted through a torque converter with a select shift type transmission with a minimum of three forward and two reverse speeds or a full hydrostatic drive with infinitely variable speed control in forward and reverse.

3.14.1.1 Drive train. The drive train shall provide power to a minimum of four (4) drive wheels and obtain 60 percent of the vehicle's weight in a drawbar pull, when on level, dry, brushed concrete. When differential-type transmissions are a part of the excavator's assemblage, a limited-slip device shall be included in the drive train. Wheel drive components shall be designed to accept the maximum deliverable torque load by the drive train without adverse affects under all pulling conditions. Hydrostatic drive train shall deliver a minimum of 20,000 ft lb to any wheel.

3.14.2 Speed. The excavator shall be capable of traveling on level pavement at a minimum speed of not less than 8 miles per hour.

3.14.3 Gradeability. The excavator shall be capable of negotiating a 30 percent grade on dry concrete or asphalt.

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3.15 Outrigger(s). When specified (see 6.2), outrigger(s) consisting of beam(s) or stabilizer attached to the undercarriage (see 3.14), shall be provided. The outrigger(s) shall be hydraulically powered, locally operated or operated from the revolving upperstructure cab (see 3.9.1). The total setting time and the total retractive time of the outrigger(s) in a normal etting position on level ground shall not exceed one minute. Outrigger beam(s) shall have sufficient strength to accept full stability loading conditions, and shall be equipped with floats or outrigger pads attached in a manner that will provide for full retraction within the undercarriage width. Outrigger(s) shall be capable of being adjusted at least four inches below ground level and three inches above ground level to compensate for uneven terrain. A device to accurately determine when the excavator is level shall be provided.

3.16 Brakes. The excavator, other than hydraulically driven types, shall be equipped with the manufacturer's standard braking system, in conformance with SAE J1152. Hydraulically driven types shall comply with the applicable stopping performance, holding performance, and parking system performance criteria of SAE J1152.

3.17 Steering. The manufacturer's standard steering system shall be provided. Hydraulic skid steering is acceptable.

3.18 Lighting. When specified (see 6.2), the excavator shall be equipped with the following as a minimum:

a. Front and rear work light(s).
b. Boom mounted work light (s).
c. Two dual stop and taillight(s), and directional turn signals.
d. Interior cab light(s).

3.19 Engine. The excavator engine shall be a commercial model diesel type having horsepower, torque, and speed characteristics to meet satisfactorily all the excavator performance requirements specified herein. The diesel engine shall start within five minutes and be ready for full load operation within fifteen minutes in any ambient temperature from +110°F to -20°F. The engine shall be equipped with an hour meter, cooling liquid temperature indicator, lubricating oil pressure indicator, liquid fuel gage, a fuel tank with sufficient capacity for eight hours normal operation, and all necessary accessories for efficient operation and control. When specified (see 6.2), the battery furnished in the excavator shall be of the maintenance free type having maintenance free characteristics listed in W-B-131.

3.19.1 Cold starting system. The excavator diesel engine shall be equipped with an engine priming system or glow plug type for cold temperature engine starting. When an ether priming system is furnished, it shall be of the measured shot type with a reservoir of not less than 12 fluid ounces.

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3.20 Tires. Tires shall be of the tube or tubeless type, of high commercial quality, and shall have individual rated load-carrying capacities equal to the maximum individual tire loading imposed by the operation in accordance with TRA recommendations. When specified (see 6.2), tires shall be foam filled.

3.21 Fire extinguisher. When specified (see 6.2), a fire extinguisher shall be provided and shall be as specified.

3.22 Lifting and tiedown attachments. When specified (see 6.2), the excavator shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II and type III of MIL-STD-209. Attachments shall be indicated by a transportation plate conforming to MIL-P-514. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the excavator when shipped.

3.23 Dozer blade. When specified (see 6.2), manufacturer's standard hydraulic operated dozer blade assembly shall be provided and be capable of operating from the excavator's hydraulic system. The hydraulic dozer blade assembly shall be equipped with quick disconnect system to allow rapid hookup or removal. Manufacturer may install the dozer blade assembly on the front or rear of the excavator's undercarriage. Operating control(s) shall be conveniently located and arranged for the operator (see 3.10).

3.24 Air transportability. When specified (see 6.2), the excavator shall be subject to air transportability requirements and shall not exceed 105 inches in width and 102 inches in height in a reduced configuration. In addition, wheeled equipment shall not exceed 13,000 lbs per single axle or 20,000 lbs per tandem axle with the fuel tank 3/4 full in a reduced configuration. Achieving a reduced configuration shall be limited to the removal or relocation of mechanically attached (non-welded) components and shall not affect the transportability of the item, including the ability to negotiate, without interference, a 15-foot ramp at an angle of 17° between two horizontal surfaces. The excavator shall be transportable under its own power with the operator in the standard seat provided. Removal, relocation, or reinstallation time of all components required to achieve the reduced cnfiguration shall not exceed four manhours. Components which require removal or relocation to achieve the reduced configuration and the removal, relocation, reinstallation process shall be described in the equipment manual(s) delivered. When delivered to the government, the item(s) shall not be in the reduced configuration.

3.25 Instruction plates. The excavator shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a

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material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter. Heavy duty decals made of material which will last and remain legible for the life of the equipment are acceptable.

3.26 Identification plate. An identification plate will be furnished by the contracting officer for each excavator. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each excavator in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.27 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.28 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.29 Servicing and adjusting. Prior to acceptance of the excavator by the Government, the contractor shall service and adjust the excavator for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.

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e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The excavator shall be conspicuously tagged to identify the lubricants and their temperature range.

3.30 Workmanship.

3.30.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.30.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.30.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.30.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one excavator when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4, and, when specified, the preproduction pack inspection of 4.7 (see 4.7 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.6, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each excavator shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. The first article shall be subjected to the tests specified in 4.4.1 through 4.4.7; and including the test specified in 4.4.8. Failure to pass any phase of these tests shall be cause for rejection.

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4.4.1 Noise level tests. Operator noise level shall be verified for conformance to 3.7.1 by testing in accordance with SAE J919. Exterior noise level shall be verified for conformance to 3.7.1 in accordance with SAE J88.

4.4.2 Hydraulic system test. The engine and all operating components shall be given a warmup operation for a continuous period of not less than 30 minutes, during which all of the functions enumerated in 3.11 shall be checked to ascertain the reliability of the hydraulic system.

4.4.3 Brake test. The brakes of the excavator shall be tested to verify conformance to 3.16. Failure of test to show full conformance shall reject the first article.

4.4.4 Road test. The excavator shall be driven on a smooth, hard surfaced road at eight miles per hour for at least thirty minutes to verify conformance to the requirements specified in 3.14.2. Excavator shall be checked for performance and handling ability.

4.4.5 Gradeability test. The excavator shall negotiate a 30 percent grade to verify conformance to 3.14.3. Failure of test to show full conformance shall reject the first article.

4.4.6 Operational tests with accessories. The excavator, together with bucket, located in rough terrain shall be tested under full load conditions to ascertain compliance with the requirements of 3.12. During these tests, observance shall be made to ascertain any instability, improper functions, leakages, and overloading in the hydraulic system of the excavator. Each accessory, furnished under the contract, shall be assembled on the boom and operated throughout its functional range under full load simulating intended service conditions. Any failure of components or distortion of parts shall be cause for rejection. The first accessory shall be tested for two hours; all other accessories shall be tested for thirty minutes each.

4.4.7 Lifting capacity and load stability tests. Load stability tests shall be conducted on the excavator in accordance with SAE J1097. The balance point capacity in the least stable direction shall be determined as specified in 3.13. Balance point capacity shall be determined without outriggers.

4.4.8 Lifting and tiedown attachments test. The excavator, when equipped with lifting and tiedown attachments, shall be tested to verify that the attachments conform to the requirements specified in 3.22.

4.5 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample, and shall be the last excavator delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

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4.6 Production unit operational test. Each excavator produced in fulfillment of a contract or order shall be completely assembled, adjusted, lubricated, and otherwise serviced for operation. The engine shall be started and subjected to a warmup period as recommended by the manufacturer. The excavator shall be given a run-in test and all controls operated a sufficient number of times to ascertain that all components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction. Failure to pass any phase of this test shall be cause for rejection.

4.7 Packaging inspection. The inspection of preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-C-3580. When specified (see 6.2), a preproduction pack shall be furnished for examination and test.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-C-3580 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The excavator is intended for use in excavating and loading loose gravel, sand, or soil.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
c. When alternating black and yellow stripes on the rear surfaces of the excavator are required (see 3.7.3).
d. When a diesel fired heater is required (see 3.9.1).
e. List of attachments and accessories for each excavator shall be specified (see 3.12).
f. When outrigger(s) are required (see 3.15).
g. When lights are required (see 3.18).
h. When battery shall be of the maintenance free type (see 3.19).
i. When foam filled tires are required (see 3.20).
j. When fire extinguisher is required (see 3.21).
k. When lifting and tiedowns are required (see 3.22).
l. When dozer blade is required (see 3.23).
m. When air transportability is required (see 3.24).
n. When color of finish coat is specified (see 3.27).
o. When a preproduction pack is required (see 4.2.1 and 4.7).
p. Level of preservation and level of packing required (see 5.1).

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6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Engine
Excavator
Hydraulic system
Noise
Safety

Custodians:

Army - AT
Navy - YD
Air Force - 99

Review Activity:

Air Force - 84

Preparing Activity:

Navy - YD

(Project 3805-0125)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-29249	EXCAVATOR, MULTIPURPOSE, WHEEL-UNDERCARRIAGE DIESEL-ENGINE-DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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